

Chapter One

Charge, Electric field & Gauss's Law

1. What is Electrostatic induction?

Electrostatic induction is a method to create or generate static electricity in a material by bringing an electrically charged object near it. This causes the electrical charges to be redistributed in the material, resulting in one side having an excess of either positive (+) or negative (−) charges.

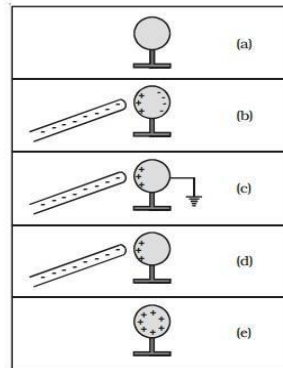


Figure 1 Electrostatic induction (Positive Charge)

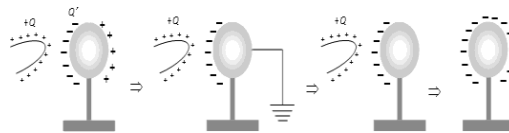


Figure 2 Electrostatic induction (Negative Charge)

2. Define: Charge, electricity.

Charge:

The fundamental and characteristics property of the particles (electron, proton etc.) that constitute matter is called charge. It is the property of material due to which, it exert a force.

Electricity:

The nature and effect of static or moving charge is called electricity.

3. What is meant by Quantization of charge?

The charge of an electron or a proton is the minimum charge available in nature. Its magnitude is $1.60218 \times 10^{-19} \text{ C}$. All other charges are the multiple of this smallest charge. It is called the quantization of charges. In nature, there is no existence of fraction of the value of e . For example, $2.5e$ or $-7.1e$ are not available in nature. Thus, If quantization is Q , we can write

$$Q = ne, \text{ where, } n = 0, 1, 2, 3, \dots$$

4. What is electric field?

Electric Field:

The region around the electric charge in which the stress or electric force act is called an electric field or electrostatic field. If the magnitude of charge is large, then it may create a huge stress around the region. The electric field is represented by the symbol E. The SI unit of the electric field is newton per coulomb which is equal to volts per meter.

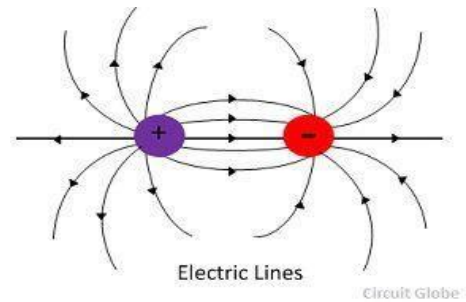


Figure 4

5. What is meant by electric field intensity?

Electric field strength or intensity:

The electric force acting on a unit charge at a point in the electric field is called electric field strength or electric field intensity.

It is denoted by E. It is a vector quantity. It can be written as, $E = F/q$

Vector form,

$$\vec{E} = \vec{F}/q$$

Where, F is the force acting on a point charge q.

The S.I. unit of electric field strength is $N C^{-1}$.

6. Write the expression for electric field intensity due to point charge?

The electric field due to a point charge:

Let a test charge is q placed at a distance r from a point charge q_0 . The magnitude of force acting on q_0 is given by

$$F = \frac{1}{4\pi \epsilon_0} \frac{qq_0}{r^2}$$

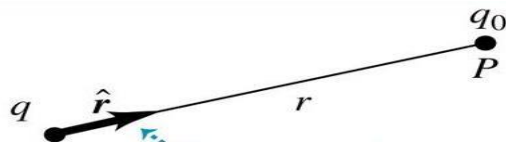


Figure 5

Hence the electric field strength at point p is

$$\vec{E} = \frac{\vec{F}}{q_0} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^3} \hat{r}$$

7. What is electric dipole?

An electric dipole is a separation of positive and negative charges. If a pair of electric charges of equal magnitude but opposite sign, separated by some distance, then the system is called an electric dipole.

8. What is meant by electric dipole moment?

The electric dipole moment is a measure of the separation of positive and negative electrical charges within a system, that is, a measure of the system's overall polarity. The SI units for electric dipole moment are coulomb-meter (C·m)

9. Calculate electric field intensity (E) for an electric dipole.

Electric Field Due to a dipole:

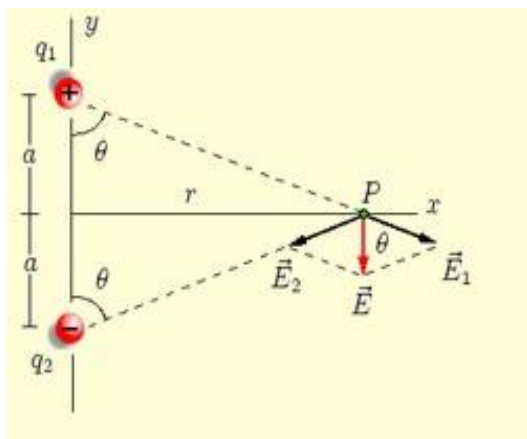


Figure 6

Figure shows an electric dipole consisting of two equal but opposite charges +q and -q which are at 2a distance apart. Now we find the expression for electric field at a point p, a distance r from the dipole. Let $\vec{E}_1$ and $\vec{E}_2$ be the electric field due to charge +q and -q respectively. Now, total electric field

$$\vec{E} = \vec{E}_1 + \vec{E}_2$$

$$\begin{aligned} \text{Where } E_1 = E_2 &= \frac{1}{4\pi\epsilon_0} \frac{q}{\left(\frac{a^2}{r^2} + 1\right)^{3/2}} \\ &= \frac{1}{4\pi\epsilon_0} \frac{q}{a^2 + r^2} \end{aligned}$$

The vector sum $\vec{E}$ of $\vec{E}_1$ and $\vec{E}_2$ points vertically downward and has the magnitude.

$$E = \sqrt{E_1^2 + E_2^2 + 2E_1E_2\cos 2\theta}$$

$$E = \sqrt{E_1^2 + E_1^2 + 2E_1E_1\cos 2\theta} \quad [\because E_1 = E_2]$$

$$E = \sqrt{2E_1^2 + 2E_1^2\cos 2\theta}$$

$$= \sqrt{2E_1^2(1 + \cos 2\theta)}$$

$$= \sqrt{2E_1^2 2\cos^2 \theta}$$

$$\sqrt{4E_1^2\cos^2 \theta}$$

$$E = 2E_1 \cos \theta$$

$$\text{From figure, } \cos \theta = \frac{a}{\sqrt{a^2 + r^2}}$$

$$\therefore E = 2E_1 \cos \theta$$

$$= \frac{2}{4\pi\epsilon_0} \frac{q}{a^2 + r^2} \frac{a}{\sqrt{a^2 + r^2}}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{2aq}{(a^2 + r^2)^{3/2}}$$

But if we assume that $r \gg a$ we get

$$E = \frac{1}{4\pi\epsilon_0} \frac{(2a)q}{r^3}$$

Again the product $(2a)q$ is called the electric dipole moment p .

$$E = \frac{1}{4\pi\epsilon_0} \frac{p}{r^3} \dots \dots \dots (1)$$

Equation (1) represents the electric field due to a point charge.

10. What is meant by charge density?

The charge density of a surface of any substance is defined as the total amount of charge existing around a point per unit area. Sometimes it is called surface charge density. Let

Q = amount of charge

A = area of cross section of substance

$$\text{Charge density, } \sigma = \frac{Q}{A}$$

Unit: Cm^{-2}

11. Write the expression of electric field intensity due to a ring of charge q and of radius a .

Electric field due to a ring of charge q and of radius a :

Let us consider a small amount of charge dq in the ring. The total electric field due to a points on the ring is E_x .

So,

$$\underline{dE_n = dE \cos \theta}$$

$$E = \frac{1}{4\pi\epsilon_0} \frac{Q}{r^2}$$

$$\begin{aligned} dE_x &= k_e \frac{dq}{r^2} \cos \theta = k_e \frac{dq}{x^2 + a^2} \cos \theta \\ &= k_e \frac{dq}{x^2 + a^2} \frac{x}{r} = k_e \frac{dq}{x^2 + a^2} \frac{x}{\sqrt{x^2 + a^2}} \\ E_x &= \int \frac{k_e x dq}{(x^2 + a^2)^{3/2}} = \frac{k_e x}{(x^2 + a^2)^{3/2}} \int dq \\ &= \frac{k_e x Q}{(x^2 + a^2)^{3/2}} \end{aligned}$$

At altitude $r_1 = 60$ km

$$E = 100 \text{ N}$$

So,

If we assume that $x \gg a$ we get

$$\begin{aligned} E &= E_x = \frac{1}{4\pi\epsilon_0} \frac{qx}{x^3} \\ \therefore E &= \frac{1}{4\pi\epsilon_0} \frac{q}{x^2} \end{aligned}$$

12. Deduce an expression of electric field intensity due to a long straight, uniformly charged wire. Or, Figure shows a section of an infinite line of a charge whose linear charge density (that is, the charge per unit length, measured in coul /meter) has the constant value λ . Calculate the field E a distance y from the line.

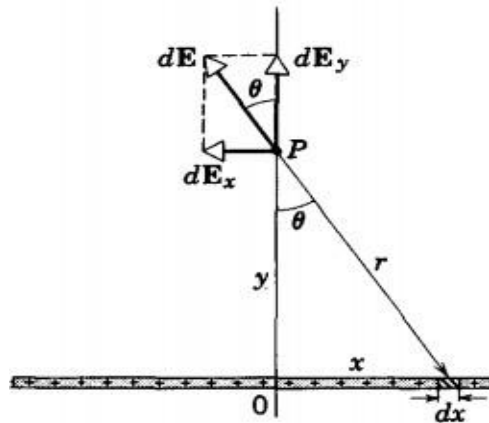


Figure 8. A section of an infinite line of charge

The figure shown a section of an infinite line of charge whose linear charge density (that is, the charge per unit length, measured in coul/meter) has the constant value λ .

The magnitude of the field contribution dE due to charge element $dq = \lambda dx$ is given by

$$\begin{aligned} dE &= \frac{1}{4\pi\epsilon_0} \frac{dq}{r^2} \\ &= \frac{1}{4\pi\epsilon_0} \frac{\lambda dx}{y^2 + x^2} \end{aligned}$$

According to the figure we can see that $dE \sin\theta$ must be zero because every charge on the right has a corresponding element on the left such that their field contributions in the x direction cancel.

$$\begin{aligned} E &= \int dE = \int_{-\alpha}^{\alpha} \cos\theta dE \\ &= \int_0^{\alpha} 2dE \cos\theta \\ &= 2 \int_0^{\alpha} \frac{1}{4\pi\epsilon_0} \frac{\lambda dx}{y^2 + x^2} \cos\theta \end{aligned}$$

According to the geometry rules we can write,

$$\begin{aligned} E &= 2 \int_0^{\alpha} \frac{1}{4\pi\epsilon_0} \frac{\lambda dx}{y^2 + x^2} \frac{y}{\sqrt{y^2 + x^2}} \\ &= 2 \int_0^{\alpha} \frac{1}{4\pi\epsilon_0} \frac{\lambda dx}{y^2 + x^2} \frac{y}{(y^2 + x^2)^{\frac{1}{2}}} \end{aligned}$$

$$= \frac{1}{4\pi\epsilon_0} \int_0^\alpha \frac{y\lambda dx}{(y^2 + x^2)^{\frac{3}{2}}}$$

$$= \frac{\lambda}{2\pi\epsilon_0} \int_0^\alpha \frac{y dx}{(y^2 + x^2)^{\frac{3}{2}}}$$

Again we know,

$$x = y \tan\theta$$

$$dx = y \sec^2\theta d\theta \quad [\text{by differentiating}]$$

$$\text{When } x = 0, \text{ then } \theta = 0$$

$$\text{When } x = \alpha, \text{ then } \theta = \frac{\pi}{2}$$

$$\therefore E = \frac{\lambda}{2\pi\epsilon_0} \int_0^{\frac{\pi}{2}} \frac{y^2 \sec^2\theta d\theta}{(y^2 + y^2 \tan^2\theta)^{\frac{3}{2}}}$$

$$E = \frac{\lambda}{2\pi\epsilon_0} \int_0^{\frac{\pi}{2}} \frac{y^2 \sec^2\theta d\theta}{\{y^2(1 + \tan^2\theta)\}^{\frac{3}{2}}}$$

$$= \frac{\lambda}{2\pi\epsilon_0} \int_0^{\frac{\pi}{2}} \frac{y^2 \sec^2\theta d\theta}{(y^2)^{\frac{3}{2}} (\sec^2\theta)^{\frac{3}{2}}}$$

$$= \frac{\lambda}{2\pi\epsilon_0} \int_0^{\frac{\pi}{2}} \frac{y^2 \sec^2\theta d\theta}{y^3 \sec^3\theta}$$

$$= \frac{\lambda}{2\pi\epsilon_0} \int_0^{\frac{\pi}{2}} \frac{d\theta}{y \sec\theta}$$

$$= \frac{\lambda}{2\pi\epsilon_0 y} \int_0^{\frac{\pi}{2}} \cos\theta d\theta$$

$$= \frac{\lambda}{2\pi\epsilon_0 y} [\sin\theta]_0^{\frac{\pi}{2}}$$

$$= \frac{\lambda}{2\pi\epsilon_0 y}$$

Which is the required expression for electric field intensity at any point due to a line of charge.

13. Expression for Torque and potential energy of an electric dipole in a uniform electric field. Or, Derive the Equation of Torque and stored energy when an electric dipole is placed in a uniform electric field.

Figure shows an electric dipole consisting of two equal but opposite charges $+q$ and $-q$ that are placed a distance $2a$ apart. This arrangement is placed in a uniform electric field $\vec{E}$. The dipole moment $p(=2aq)$ makes an angle θ with $\vec{E}$. The two forces $+\vec{F}$ and $-\vec{F}$ act as shown in figure where

$$F = qE$$

So, net force is zero but there is a net torque about an axis through O which is given by

$$\tau = 2F(asin\theta)$$

$$\tau = 2aFsin\theta$$

$$= 2aqE sin\theta$$

$$= PE sin\theta$$

$$\therefore \vec{\tau} = \vec{P} \times \vec{E}$$

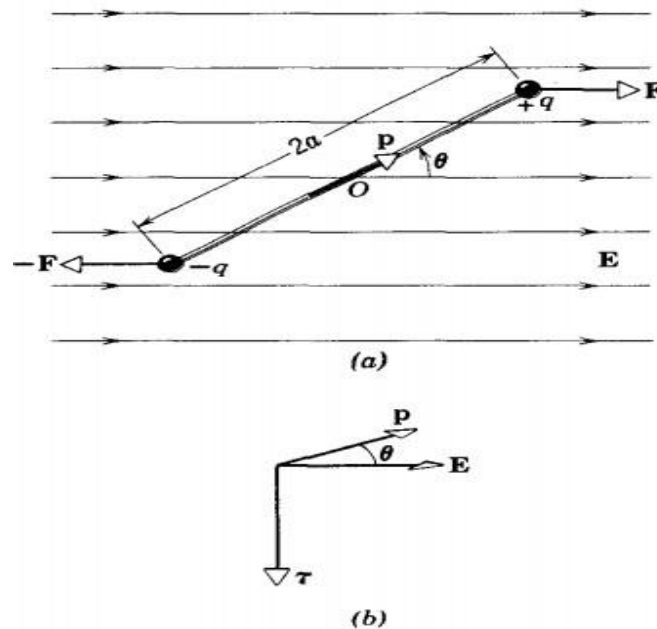


Figure 9 (a) An electric dipole in a uniform external field (b) An oblique view, illustrating

$$\vec{\tau} = \vec{P} \times \vec{E}$$

Now work must be done to change the orientation of an electric dipole by an external agent and this work is stored as potential energy.

If θ has an initial value of θ_0 , The work required to change the dipole axis to an angle θ is given by

$$W = \int_{\theta_0}^{\theta} dw = \int_{\theta_0}^{\theta} \tau d\theta = U$$

$$\begin{aligned} U &= \int_{\theta_0}^{\theta} PE \sin\theta d\theta \\ &= PE [-\cos\theta]_{\theta_0}^{\theta} \\ &= PE(-\cos\theta + \cos\theta_0) \end{aligned}$$

If we choose reference orientation θ_0 to have any convenient value $\theta_0 = 90^\circ$, we have

$$U = -PE \cos\theta$$

$$\therefore U = -\vec{p} \cdot \vec{E}$$

Which is the required expression for potential energy of a dipole in a uniform electric field.

14. What is line of forces?

Line of forces: If a unit positive charge is placed in an electric field then the path through the charge moves are called electric lines of forces of that electric field.

15. Properties of electric lines of force.

Properties of electric lines of force:

- Electric lines of force start from positive charge and end at the negative charge.
- Electric lines of force originate and finish perpendicularly to the surface.
- Electric lines of force never intersect each other.
- Electric lines of force repulse one another.
- Electric lines of force do not pass through a conductor.
- When the charges are positive the electric field lines are directed outward from the charges, when the charges are negative the electric field lines are directed inward toward the charge, when both positive and negative charges are present in the electric field, electric field lines generated from the positive charge are terminated at the negative charges. When both charges are the same the mutual electric field lines are as shown in figure.

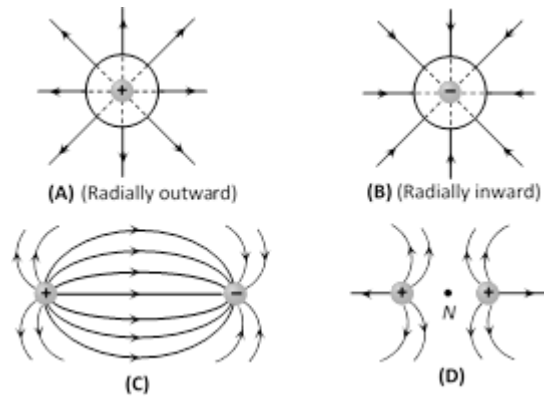


Figure 10. Electric lines of force

16. Define: Electric Flux, Electric Potential.

Electric flux: The number of electric lines of force or field lines passing through a pale or surface is called electric flux. It is denoted by Φ_E . Flux is property of any vector field. It refers to a hypothetical surface in the field, which may be open or closed. For closed surface, Φ_E is positive if the lines of force point outward and it is negative for that of inward.

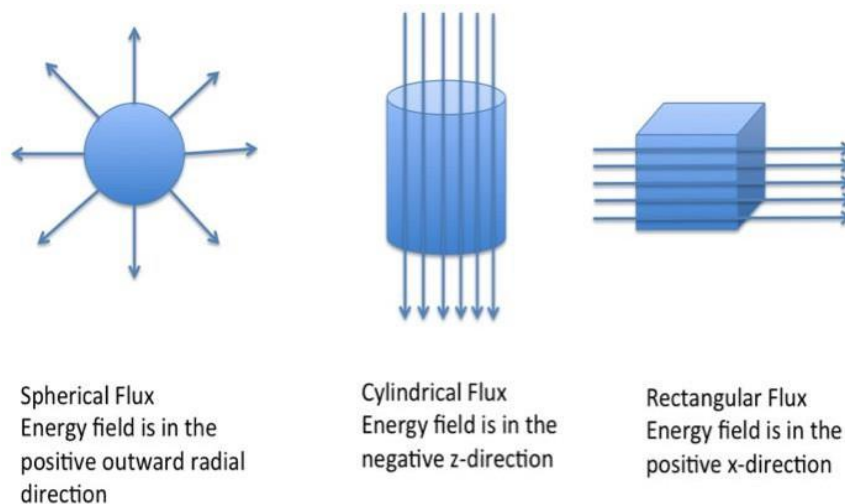


Figure 11 Electric flux

Electric potential:

The energy required to bring unit charge from infinity to a point in the electric field is known as electric potential.

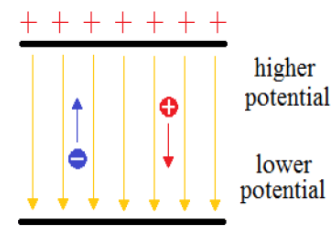


Figure 12

17. State and explain Coulomb's law.

Coulomb's Law:

In 1785 Charles Augustin de Coulomb first measured electrical attractions and repulsions quantitatively and deduced a law that governs them which is known as Coulomb's law. The law can be stated as below:

“The attractive or repulsive force between two charges is directly proportional to the product of two charges and inversely proportional to the square of their distance and the force acts along their joining line.

Explanation:

Let q_1 and q_2 be two point charges and the distance between them is r . If they exert a force of magnitude F , then according to Coulomb's law,

$$F \propto q_1 q_2 \text{ When, } r \text{ is constant and}$$

$$F \propto \frac{1}{r^2} \text{ When } q_1 \text{ and } q_2 \text{ are constant}$$

When r , q_1 and q_2 all vary, then

$$F \propto \frac{q_1 q_2}{r^2}$$

$$\text{or, } F = k \frac{q_1 q_2}{r^2}$$

In vector form,

$$\vec{F} = k \frac{q_1 q_2}{r^2} \hat{r}$$

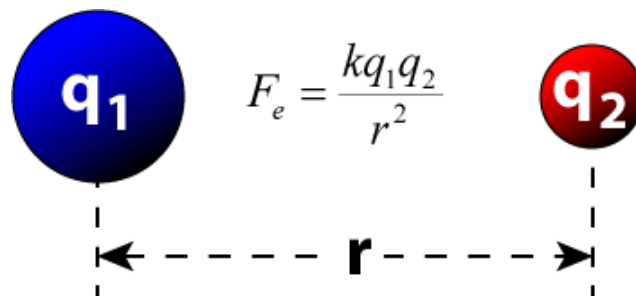


Figure 3

Here, k is a proportionality constant. Its value depends on the nature of the medium and the units of q_1 , q_2 , F and r .

In S. I. units,

$$k = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ nm}^2\text{C}^{-2}$$

The quantity ϵ_0 is called the permittivity constant, sometimes appears separately in equations and is

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2/\text{N.m}^2$$

18. State and explain Gauss's law.

Gauss's Law

Eminent mathematician Karl F. Gauss gave this law.

The total charge enclosed by closed surface is equal to ϵ_0 times the electric flux with the closed surface.

If the area of a closed surface is S and the total charge enclosed by that surface is q , then according to Gauss's law,

$$\epsilon_0 \Phi = q$$

$$\text{Or, } \epsilon_0 \oint \vec{E} \cdot d\vec{S} = q$$

Here, ϵ_0 is the permittivity of vacuum.

19. What is meant by Gaussian surface?

Gaussian surface

The imaginary closed surface having a collection of positive and negative charges which is used to explain the Gauss's law is called the Gaussian surface.

20. Deduce Coulomb's law from Gauss's law.

Coulomb's law from Gauss's law:

Let us consider an isolated point charge q surrounded by a spherical Gaussian surface of radius r .

Now, $\vec{E}$ must be normal to the surface and must have some magnitude for all points on the surface.

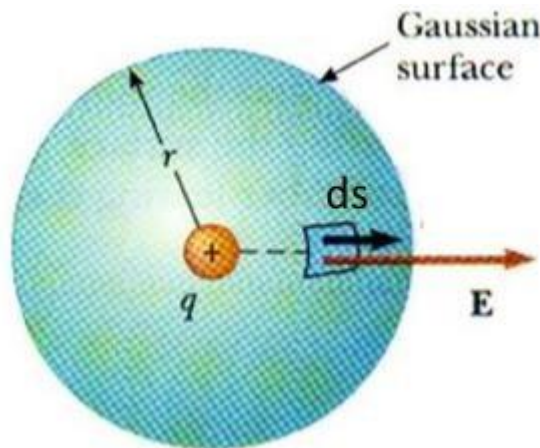


Figure 13

According to Gauss's law,

$$\epsilon_0 \oint \vec{E} \cdot d\vec{S} = q$$

Since, $\vec{E}$ and $d\vec{S}$ at any point on the Gaussian surface are directed outward, so the angle between them is Zero.

$$\therefore \epsilon_0 \oint E \cdot ds \cos\theta = q$$

$$\text{or, } \epsilon_0 \oint E \cdot ds \cos 0^\circ = q$$

$$\text{or, } \epsilon_0 \oint E \cdot ds = q$$

$$\text{or, } \epsilon_0 E \oint ds = q$$

$$\text{or, } \epsilon_0 E (4\pi r^2) = q$$

Where, $\oint ds = 4\pi r^2$ is the area of the surface.

$$E = \frac{1}{4\pi \epsilon_0} \frac{q}{r^2} \dots\dots\dots (1)$$

Equation (1) gives the magnitude of $\vec{E}$ at any point a distance r from an isolated point charge q .

Let us put a second point charge q_0 at the point at where $\vec{E}$ is calculated. The magnitude of the force that acts on it is

$$\begin{aligned} F &= q_0 E \\ &= \frac{1}{4\pi \epsilon_0} \frac{qq_0}{r^2} \\ \therefore F &= \frac{1}{4\pi \epsilon_0} \frac{qq_0}{r^2} \end{aligned}$$

Which is precisely coulomb's law. Thus we have deduced coulomb's law from Gauss's law.

21. Write down the Application of Gauss's law.

Application of Gauss's law:

Although Gauss's law is always true it is only a useful tool to calculate the electric field if the charge distribution is symmetric:

1. If the charge distribution has **spherical symmetry**, then Gauss's law can be used with concentric spheres as Gaussian surfaces.
2. If the charge distribution has **cylindrical symmetry**, then Gauss's law can be used with coaxial cylinders as Gaussian surfaces.
3. If the charge distribution has **plane symmetry**, then Gauss's law can be used with pill boxes as Gaussian surfaces.

22. An infinite length rod have linear charge density λ . Calculate E for the rod at a distance 'r' from the rod.

Or, Find an expression of $\vec{E}$ for a rod (cylindrical surface).

Let us consider a section of an infinite rod with uniformly charge density λ , is constant for all points on the line.

Let the rod be surrounded by a cylindrical Gaussian surface of radius r and length h.

If $\vec{E}$ is the electric field, according to Gauss's law we can write

$$\epsilon_0 \oint \vec{E} \cdot d\vec{S} = \lambda h$$

$$\text{or, } \epsilon_0 \oint E \cdot ds = \lambda h$$

$$\text{or, } \epsilon_0 E \oint ds = \lambda h$$

$$\text{or, } \epsilon_0 E (2\pi r h) = \lambda h$$

$$\therefore E = \frac{\lambda}{2\pi \epsilon_0 r}$$

The direction of $\vec{E}$ is radially outward for a line of positive charge.

In vector form,

$$\vec{E} = \frac{\lambda r \rightarrow}{2\pi \epsilon_0 r^2}$$

23. What is electric potential energy?

Electric potential:

The work done in bringing a positive charge from infinity up to a point is called the electric potential at that point.

If the work done is W and the charge is q_0 , the electric potential V , can be written as

$$V = W/q_0$$

24. What do you understand by equipotential surface?

Equipotential surface:

An equipotential surface is a three dimensional surface on which the electric potential is the same at every point.

25. What is meant by potential difference?

The potential difference between two points can be defined as the work done W_{AB} in moving a test charge q_0 from a point A to a point B keeping the charge in equilibrium. Thus,

$$V_B - V_A = \frac{W_{AB}}{q_0}$$

$$\text{Unit: Volt} = \frac{\text{Joul}}{\text{coulomb}}$$

27. Derive an expression for potential due to point charge.

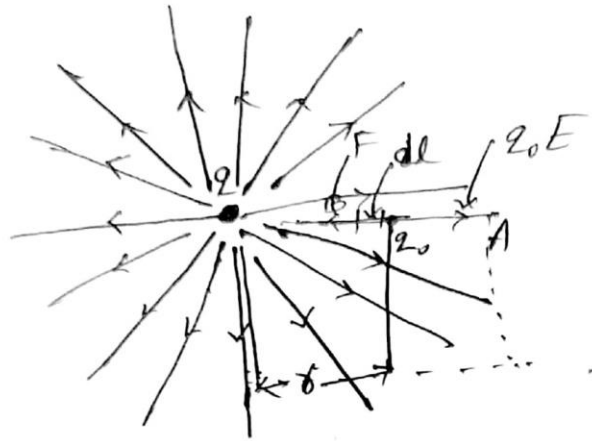


Fig: A test charge q_0 is moved A to B in the field set up by a point charge Q .

Let us consider A , B and Q lie on a straight line. Let us compute the potential difference between points A and B , assuming that a test charge q_0 is moved without acceleration, along a radial line from A to B .

$\vec{E}$ point to the right and $d\vec{l}$ to the left.

$$\text{Therefore, } \vec{E} \cdot d\vec{l} = E dl \cos 180^\circ \\ = -E dl$$

As we move a distance dl to the left, we are moving in the direction of decreasing r .

Thus.

$$dl = -dr$$

$$\begin{aligned}\text{So, } \vec{E} \cdot d\vec{l} &= -E dl \\ &= -E(-dr) \\ &= E dr\end{aligned}$$

The potential difference between A and B

$$\begin{aligned}V_B - V_A &= - \int_A^B \vec{E} \cdot d\vec{l} \\ &= - \int_{\pi_A}^{\pi_B} E dr\end{aligned}$$

$$V_B - V_A = - \int_{\pi_A}^{\pi_B} \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \cdot dr$$

$$= - \int_{\pi_A}^{\pi_B} \frac{q}{4\pi\epsilon_0} \frac{dr}{r^2}$$

$$= - \frac{q}{4\pi\epsilon_0} \left[\frac{r^{-2+1}}{-2+1} \right]_{\pi_A}^{\pi_B}$$

$$= - \frac{q}{4\pi\epsilon_0} \left[\frac{r^{-1}}{-1} \right]_{\pi_A}^{\pi_B}$$

$$= \frac{q}{4\pi\epsilon_0} \left[\frac{1}{r} \right]_{\pi_A}^{\pi_B}$$

$$= \frac{q}{4\pi\epsilon_0} \left[\frac{1}{\pi_B} - \frac{1}{\pi_A} \right]$$

At $\pi_A \rightarrow \infty$, $V_A = 0$,

$$\text{then } V_B - V_A = V = \frac{1}{4\pi\epsilon_0} \frac{q}{\pi_B} = \frac{1}{4\pi\epsilon_0} \frac{q}{\pi}$$

$$\therefore V = \frac{1}{4\pi\epsilon_0} \frac{Q}{r}$$

28. Derive an expression for electric potential due to a charged disk.

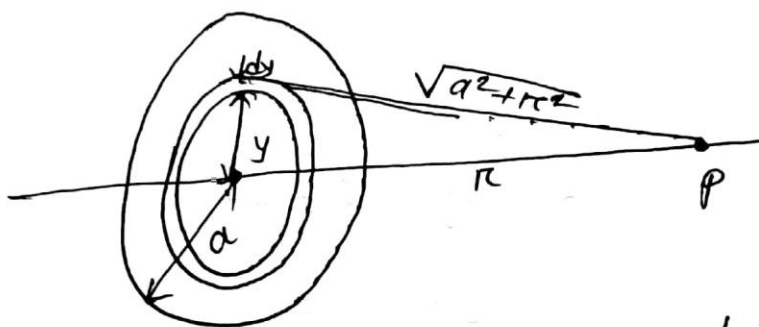


Fig: 28. A uniformly charged circular disk of radius a .

$$\text{Area of the strip} = 2\pi y \times dy = 2\pi y dy$$

$$\text{charge density, } \sigma = \frac{dQ}{2\pi y dy}$$

$$\therefore \text{charge } dQ = 2\pi y dy \times \sigma$$

Now, the potential at the point P due to the strip is,

$$dV = \frac{1}{4\pi\epsilon_0} \frac{dQ}{\sqrt{a^2 + r^2}}$$

$$\Rightarrow dV = \frac{1}{4\pi\epsilon_0} \frac{2\pi y dy \times \sigma}{\sqrt{a^2 + r^2}}$$

$$\Rightarrow dV = \frac{1}{4\pi\epsilon_0} \frac{\sigma(2\pi y)(dy)}{\sqrt{\pi^2 + y^2}}$$

$$= \frac{\sigma}{2\pi\epsilon_0} \frac{1}{\sqrt{\pi^2 + y^2}} y dy$$

The potential V is found by integrating the above eqⁿ from 0 to a is given by

$$\therefore V = \int dV = \int_0^a \frac{\sigma}{2\epsilon_0} \frac{1}{\sqrt{\pi^2 + y^2}} y dy$$

$$= \frac{\sigma}{2\epsilon_0} \int_0^a \frac{y dy}{\sqrt{\pi^2 + y^2}}$$

Let

$$\pi^2 + y^2 = z^2$$

$$2y dy = 2z dz$$

$$y dy = z dz$$

$$\therefore V = \frac{\sigma}{2\epsilon_0} \int_0^a \frac{z dz}{z}$$

$$= \frac{\sigma}{2\epsilon_0} \int_0^a dz = \frac{\sigma}{2\epsilon_0} [z]_0^a$$

$$= \frac{\sigma}{2\epsilon_0} [\sqrt{\pi^2 + y^2}]_0^a$$

$$\therefore V = \frac{\sigma}{2\epsilon_0} [\sqrt{\pi^2 + a^2} - \pi] \dots \dots \dots (1)$$

This general result is valid for all values of π .

In the special case if $\pi \gg a$, the

$$\sqrt{\pi^2 + a^2} = (\pi^2 + a^2)^{\frac{1}{2}}$$

$$= \left\{ \pi^2 \left(1 + \frac{a^2}{\pi^2} \right) \right\}^{\frac{1}{2}}$$

$$= \pi \left(1 + \frac{a^2}{\pi^2} \right)^{\frac{1}{2}}$$

$$= \pi \left\{ 1 + \frac{1}{2} \frac{a^2}{\pi^2} + \frac{\frac{1}{2}(\frac{1}{2}-1)}{2!} \left(\frac{a^2}{\pi^2} \right)^2 + \dots \right\}$$

$$\left[\because (1+x)^n = 1 + nx + \frac{n(n-1)}{2!} x^2 + \dots \right]$$

$$\approx \pi \left(1 + \frac{1}{2} \frac{a^2}{\pi^2} \right)$$

$$= \pi + \frac{1}{2} \frac{a^2}{\pi}$$

$$\therefore V = \frac{\sigma}{2\epsilon_0} \left(\pi + \frac{1}{2} \frac{a^2}{\pi} - \pi \right)$$

$$= \frac{\sigma a^2}{4\epsilon_0 \pi}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{\sigma \pi a^2}{\pi}$$

$$\therefore \boxed{V = \frac{1}{4\pi\epsilon_0} \frac{q}{\pi}}$$

$$\boxed{\begin{aligned} \sigma &= \frac{q}{A} = \frac{q}{\pi a^2} \\ \therefore q &= \sigma \pi a^2 \end{aligned}}$$

29. Derive an expression for potential due to a dipole.

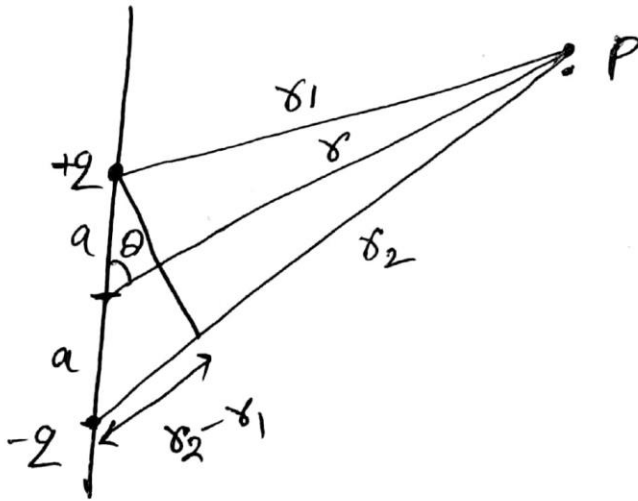


Figure 29: An electric dipole with charge q .

Figure 29, shows an ~~potential~~ Electric dipole of charge q , separated by a $2a$ distance apart.

Now, the potential at point P is

$$\begin{aligned}
 V &= \sum_{i=1}^n V_i = V_1 + V_2 \\
 &= \frac{1}{4\pi\epsilon_0} \frac{q}{r_1} - \frac{1}{4\pi\epsilon_0} \frac{q}{r_2} \\
 &= \frac{q}{4\pi\epsilon_0} \left(\frac{1}{r_1} - \frac{1}{r_2} \right) \\
 V &= \frac{q}{4\pi\epsilon_0} \left(\frac{r_2 - r_1}{r_1 r_2} \right) \dots \dots (1)
 \end{aligned}$$

When $r \gg 2a$

$$\cos \theta = \frac{r_2 - r_1}{2a}$$

$$\therefore r_2 - r_1 = 2a \cos \theta$$

$$\text{and } r_1 r_2 = r^2$$

Thus,

$$V = \frac{q}{4\pi\epsilon_0} \frac{2a \cos \theta}{r^2}$$
$$= \frac{1}{4\pi\epsilon_0} \frac{(2aq) \cos \theta}{r^2}$$

$$\therefore V = \frac{1}{4\pi\epsilon_0} \frac{p \cos \theta}{r^2}$$

$$\left[\text{where, } p = 2aq = \text{dipole moment} \right]$$

which is required solution.

30. What do understand by Capacitance?

The amount of charge required to increase the potential of a conductor by one unit is called the capacitance of the conductor.

Its unit is Farad (F)

Mathematically $C = \frac{Q}{V}$

31. What is capacitor?

The mechanical device in which an insulator is placed between two conductors in order to store charge is called capacitor.

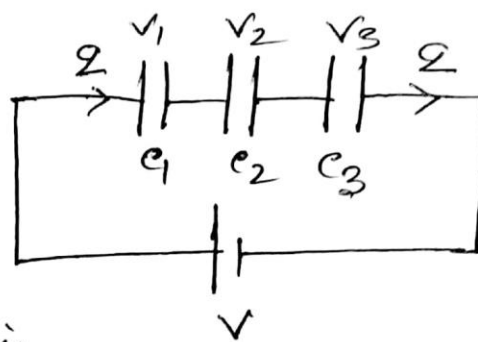
32. What is Parallel plate Capacitor?

If ~~the~~ two identical metallic plates of some surface area are placed parallel to each other and one is connected with a generator and the other is connected to the earth and a dielectric substance is inserted between the plates then the produced device is called parallel plate capacitor.

33. Calculate Equivalent Capacitance for Series and Parallel combination.

Calculation series combination:

For series combination charge is same for all capacitors.



Let equivalent voltage is

$$V = V_1 + V_2 + V_3$$

$$\Rightarrow \frac{Q}{C} = \frac{Q}{C_1} + \frac{Q}{C_2} + \frac{Q}{C_3}$$

$$\therefore \boxed{\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3}}$$

We know,

$$C = \frac{Q}{V}$$
$$\therefore V = \frac{Q}{C}$$

For parallel combination:

In parallel connection voltage is same for all capacitors.

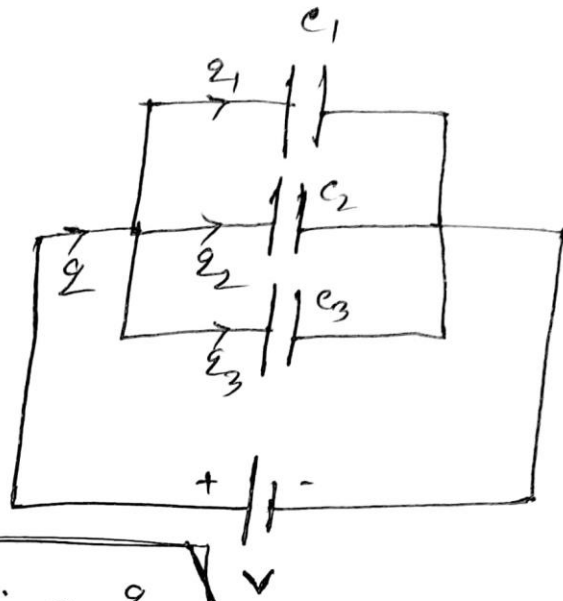
let,

$$Q = Q_1 + Q_2 + Q_3$$

$$\Rightarrow CV = C_1V + C_2V + C_3V$$

$$\begin{aligned} \therefore C &= \frac{Q}{V} \\ \therefore Q &= CV \end{aligned}$$

$$\therefore \boxed{C_p = C_1 + C_2 + C_3}$$



34. What is dielectric?

The insulator materials which are poor conductors of electricity and can sustain the force of electric fields passing through them are called dielectrics.

Dielectric materials are used to increase the capacitance of capacitors by placing them between the plates or shells. Example: ~~Glass~~ Glass, mica, mineral oils, wax etc.

25. What do you understand by dielectric constant (K)?

If the capacitance with an insulating material between the plates is C and without an insulating material is C_0 , then it is found that the ratio of the capacitances in these two cases is always a constant. This constant is called the dielectric constant, K , or relative permittivity, ϵ_r , of this insulating medium. So

$$\epsilon_r = K = \frac{C}{C_0}$$

36. What do you mean by three electric Vector?

A parallel-plate capacitor containing dielectric as

$$\begin{aligned}\frac{Q}{K\epsilon_0 A} &= \frac{Q}{\epsilon_0 A} - \frac{Q'}{\epsilon_0 A} \\ \Rightarrow \frac{Q}{\epsilon_0 A} - \frac{Q'}{\epsilon_0 A} &= \frac{Q}{K\epsilon_0 A} \\ \Rightarrow \frac{Q}{\epsilon_0 A} &= \frac{Q}{K\epsilon_0 A} + \frac{Q'}{\epsilon_0 A} \\ \Rightarrow \frac{Q}{A} &= \epsilon_0 \left(\frac{Q}{K\epsilon_0 A} \right) + \frac{Q'}{A} \quad \dots \dots \dots (1)\end{aligned}$$

Hence,

$$\frac{Q}{A} = D = \text{Free charge}$$

$$\frac{Q}{K\epsilon_0 A} = E = \text{Electric field}$$

$$\frac{Q'}{A} = P = \text{polarization charge}$$

Now, from equation (1)

$$D = \epsilon_0 E + P$$

In vector form

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P}$$

$\vec{D}$, $\vec{E}$ and $\vec{P}$ are called three electric vectors.

37. Deduce the expression for energy stored in an electric field.

Or,

38. Deduce the expression for energy stored of a charged Capacitor.

Let us consider, at a time dt , a charge dq is transferred from one place to other,

So the work done

$$dW = V dq$$

Now, the total work done

$$W = \int dW = \int_0^q V dq$$

So, the total energy storage is $U = \int_0^q V dq$

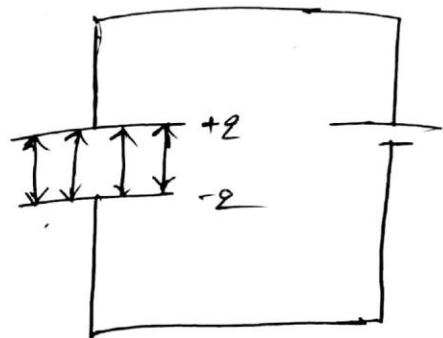


Fig: Energy storage in an electric field.

$$U = \int_0^2 \sqrt{q} dq$$

$$= \int_0^2 \frac{q}{c} dq$$

$$= \frac{1}{c} \int_0^2 q dq$$

$$= \frac{1}{c} \left[\frac{q^2}{2} \right]_0^2$$

$$U = \frac{q^2}{2c} = \frac{(ev)^2}{2c} \quad [q = ev]$$

$$= \frac{e^2 v^2}{2c}$$

$$\therefore \boxed{U = \frac{1}{2} e v^2}$$

Thus the energy density or energy per unit volume is

$$u = \frac{U}{V} = \frac{U}{Ad}$$

$$= \frac{\frac{1}{2} e v^2}{Ad}$$

$$= \frac{\frac{1}{2} \left(\frac{\epsilon A}{d} \right) (Ed)^2}{Ad}$$

$$\left[\because V = Ed \text{ and } c = \frac{\epsilon A}{d} \right]$$

$$u = \frac{1}{2} \epsilon E^2$$

$$u = \frac{1}{2} K \epsilon_0 E^2 \quad [\epsilon = \epsilon_0 K]$$

In air $K=1$,

$$\boxed{u = \frac{1}{2} \epsilon_0 E^2}$$

which is required expression.

Q9. Calculate Capacitance for parallel plate capacitor (With and Without Dielectric constant).

Let us consider a parallel plate capacitor having,

A = area of each plate of the capacitor.

d = distance between two plates.

ϵ = permittivity of the medium between plates.

Q = charge at each plate.

V = potential difference between the plates.

$\sigma = \frac{Q}{A}$ charge density at each plate.

Hence, the capacitance of the capacitor is

$$C = \frac{Q}{V} \dots \dots \dots (1)$$

Again the electric field intensity is

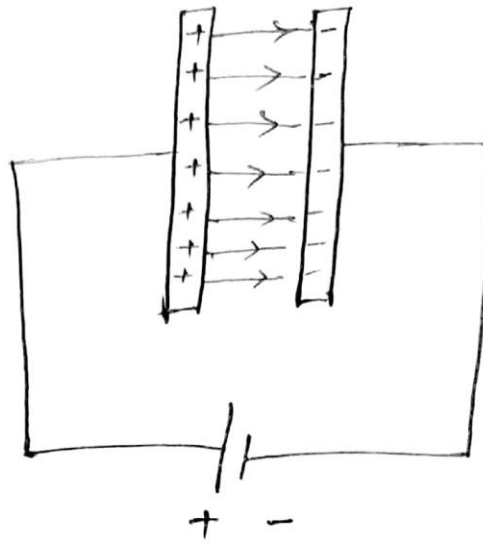
$$E = \frac{\sigma}{\epsilon_0}$$

$$\text{or, } E = \frac{Q}{A\epsilon_0} \quad [\because \sigma = \frac{Q}{A}]$$

$$\text{But, } V = Ed$$

$$V = \frac{Qd}{A\epsilon_0}$$

putting this value in eq (1)



$$C = \frac{Q}{\frac{Qd}{\epsilon A}}$$

$$\Rightarrow C = \frac{\cancel{Q} \epsilon A}{\cancel{Q} d}$$

$$\therefore C = \frac{\epsilon A}{d}$$

Since the dielectric constant,

$$K = \frac{\epsilon}{\epsilon_0}$$

$$\text{or } \epsilon = K \epsilon_0$$

$$\therefore \boxed{C = \frac{\epsilon_0 K A}{d}}$$

In air, $K = 1$, i.e. without dielectric constant,

$$\boxed{C = \frac{\epsilon_0 A}{d}}$$

which is required capacitance for ~~at~~ parallel plate capacitor.